

Roots (Part II) Homework

Name
Date
Course

* Assume that all variables are positive.

Rationalize the denominators of the following fractions.

Answers

$$\textcircled{1} \frac{2}{\sqrt{5}-6}$$

$$\frac{2\sqrt{5}+12}{-31}$$

$$\textcircled{2} \frac{7}{\sqrt{10}+\sqrt{3}}$$

$$\sqrt{10}-\sqrt{3}$$

$$\textcircled{3} \frac{\sqrt{6}-4}{\sqrt{3}-1}$$

$$\frac{3\sqrt{2}+\sqrt{6}-4\sqrt{3}-4}{2}$$

$$\textcircled{4} \frac{8+\sqrt{2}}{\sqrt{7}+2}$$

$$\frac{8\sqrt{7}-16+\sqrt{14}-2\sqrt{2}}{3}$$

$$\textcircled{5} \frac{\sqrt{11}+3\sqrt{3}}{\sqrt{5}-\sqrt{6}}$$

$$-\sqrt{55}-\sqrt{66}-3\sqrt{15}-9\sqrt{2}$$

$$\textcircled{6} \frac{7}{\sqrt{2x}}$$

$$\frac{7\sqrt{2x}}{2x}$$

$$\textcircled{7} \frac{1}{\sqrt{15x}}$$

$$\frac{\sqrt{15x}}{15x}$$

$$\textcircled{8} \frac{-4}{3\sqrt{x}}$$

$$\frac{-4\sqrt{x}}{3x}$$

$$\textcircled{9} \frac{3}{8\sqrt{11x}}$$

$$\frac{3\sqrt{11x}}{88x}$$

$$\textcircled{10} \frac{9}{\sqrt{54x^4}}$$

$$\frac{\sqrt{6}}{2x^2}$$

$$\textcircled{11} \frac{1}{\sqrt{20x^3}}$$

$$\frac{\sqrt{5x}}{10x^2}$$

$$\textcircled{12} \sqrt{\frac{10}{49x^8}}$$

$$\frac{\sqrt{10}}{7x^4}$$

$$\textcircled{13} \sqrt{\frac{3}{64x^{10}}}$$

$$\frac{\sqrt{3}}{8x^5}$$

$$\textcircled{14} \sqrt{\frac{9}{7x}}$$

$$\frac{3\sqrt{7x}}{7x}$$

$$\textcircled{15} \sqrt{\frac{4}{x}}$$

$$\frac{2\sqrt{x}}{x}$$

$$\textcircled{16} \sqrt{\frac{3}{10x}}$$

$$\frac{\sqrt{30x}}{10x}$$

$$\textcircled{17} \sqrt{\frac{17x}{2}}$$

$$\frac{\sqrt{34x}}{2}$$

$$\textcircled{18} \sqrt{\frac{63x^4}{10}}$$

$$\frac{3x^2\sqrt{70}}{10}$$

$$\textcircled{19} \sqrt{\frac{56x^9}{5}}$$

$$\frac{2x^4\sqrt{70x}}{5}$$

$$\textcircled{20} \sqrt{\frac{10}{27x^6}}$$

$$\frac{\sqrt{30}}{9x^3}$$

$$\textcircled{21} \sqrt{\frac{3}{8x^5}}$$

$$\frac{\sqrt{6x}}{4x^3}$$

$$\textcircled{22} \sqrt{\frac{32}{99x^7}}$$

$$\frac{4\sqrt{22x}}{33x^4}$$

$$\textcircled{23} \sqrt{\frac{28x^2}{75}}$$

$$\frac{2x\sqrt{21}}{15}$$

$$\textcircled{24} \sqrt{\frac{50}{27x^5}}$$

$$\frac{5\sqrt{6x}}{9x^3}$$

$$\textcircled{25} \frac{\sqrt{18x^3}}{\sqrt{32x^9}}$$

$$\frac{3}{4x^3}$$

$$\textcircled{26} \frac{\sqrt{20x^{11}}}{\sqrt{5x}}$$

$$2x^5$$

$$\textcircled{27} \frac{2}{\sqrt{x}+3}$$

$$\frac{2\sqrt{x}-6}{x-9}$$

$$\textcircled{28} \frac{1}{4-\sqrt{10x}}$$

$$\frac{4+\sqrt{10x}}{16-10x}$$

$$\textcircled{29} \frac{\sqrt{7x}-8}{3-\sqrt{5}}$$

$$\frac{3\sqrt{7x}+\sqrt{35x}-24-8\sqrt{5}}{4}$$

$$\textcircled{30} \frac{\sqrt{x}+2}{\sqrt{3x}-1}$$

$$\frac{x\sqrt{3}+\sqrt{x}+2\sqrt{3x}+2}{3x-1}$$